Digital Cloak

Orange Pi 5 Plus Setup & Configuration Guide

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Software

- Central Kismet Server
- ELK Stack Server
- SD Card With Operating System (Armbian):
  - Ensure that you select the Desktop version of Armbian (left option).
  - Armbian Operating System Download Website

Peripherals:

- Monitor
- HDMI Cable
- Keyboard
- Mouse

Overview

Required Tools Overview

In this section, we detail the essential hardware and software tools needed for setting up and configuring the Orange Pi 5 Plus. We cover the necessary components, such as the Orange Pi 5 Plus itself, its compatible peripherals, and the software environment required for the setup. This guide includes links to sources where you can purchase the hardware components and download the required software.

Assembly Overview

This section provides a step-by-step guide to assembling the Orange Pi 5 Plus. From installing the eMMC module and heat sinks to connecting the WiFi card and NVME SSD, each step is described in detail with supporting images. We explain the significance of each component and offer visual aids to ensure correct installation. By following these instructions, you can ensure your Orange Pi 5 Plus is correctly assembled and ready for the startup process.

Startup Overview

Here, we outline the procedures for powering on and configuring your Orange Pi 5 Plus for the first time. This includes loading the operating system onto an SD card, connecting peripherals, and ensuring proper power and data connections. We also cover the critical steps of flashing the eMMC, assigning hostnames, and verifying that all hardware components are functioning correctly.

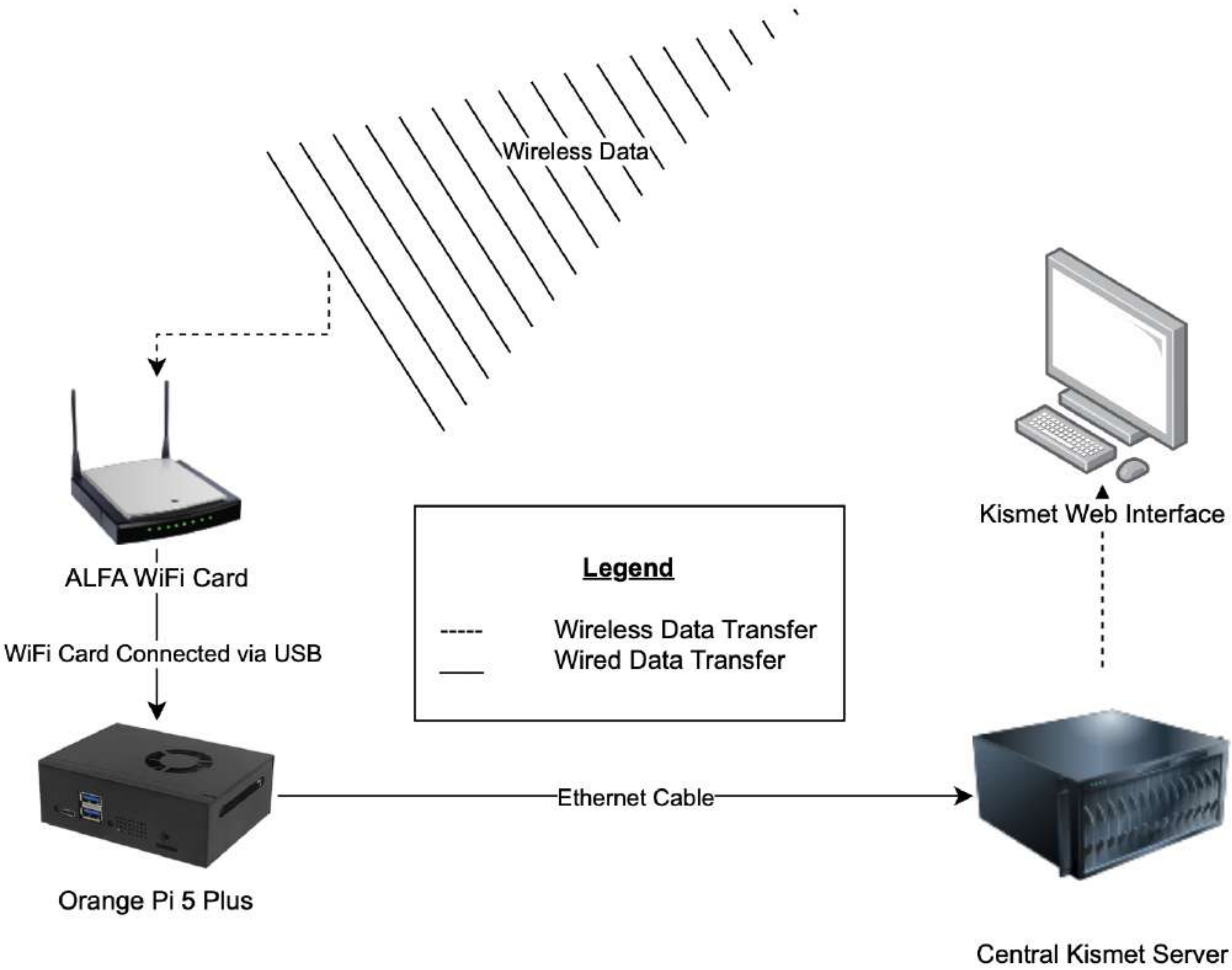
Kismet Overview

This section guides you through the configuration and setup of Kismet, a wireless network detector, sniffer, and intrusion detection system. From initial server configuration to starting data capture using WiFi and RTL-SDR interfaces, we provide detailed commands and instructions. This ensures that your Orange Pi 5 Plus can effectively monitor wireless networks and capture data for analysis.

Metricbeat Overview

Metricbeat is used to ship metrics from the Orange Pi 5 Plus to an ELK Stack for monitoring and analysis. In this section, we provide step-by-step instructions on installing Metricbeat, configuring it to communicate with your ELK server, and ensuring it runs as a service. We also include troubleshooting tips and commands to verify that Metricbeat is correctly sending data to your monitoring setup.

Information Flow Chart



Assembly

Step 1 - Install eMMC

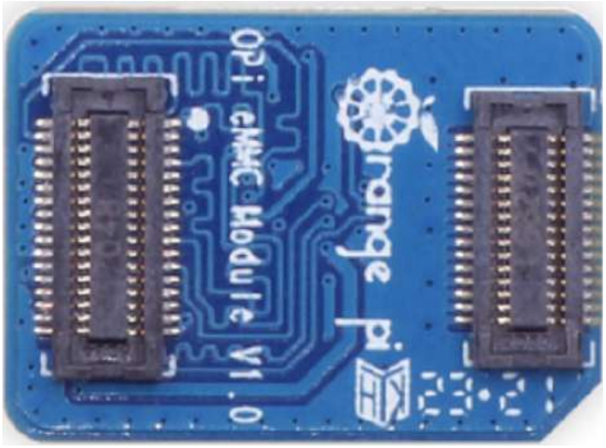
What is it?

An eMMC (embedded MultiMediaCard) is a non-volatile memory used for storage in electronic devices, including single-board computers like the Orange Pi 5 Plus.

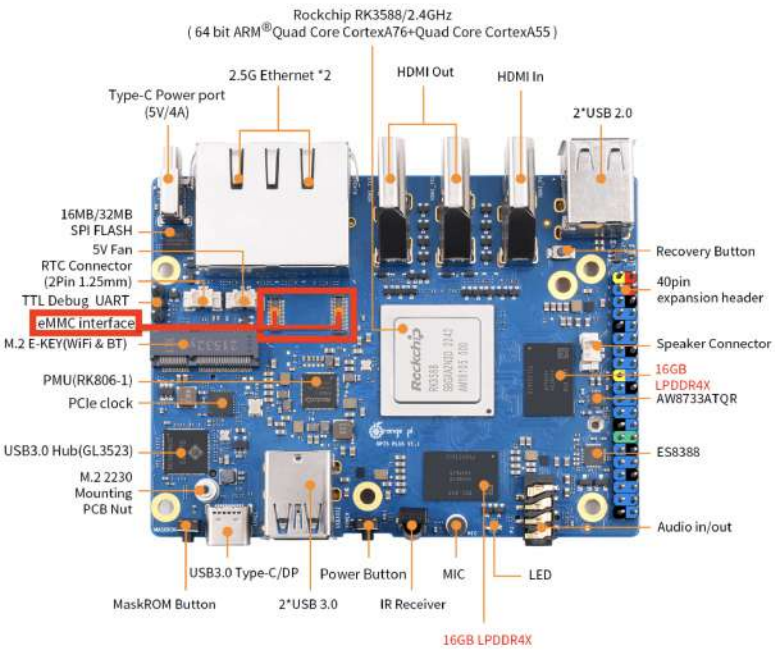
Why is an eMMC important?

We are utilizing this type of storage to load and boot the contents of the SD Card. This is safer, faster, and more reliable than booting from an SD Card.

Connect 256GB eMMC Module to the Orange Pi motherboard:



- Ensure the eMMC is oriented in the correct location.
- The chipped edge should be facing in the direction of the wireless PCIe clock.
- The eMMC is installed after an audible click is made.



Step 2 - Install Heat Sinks

What are they?

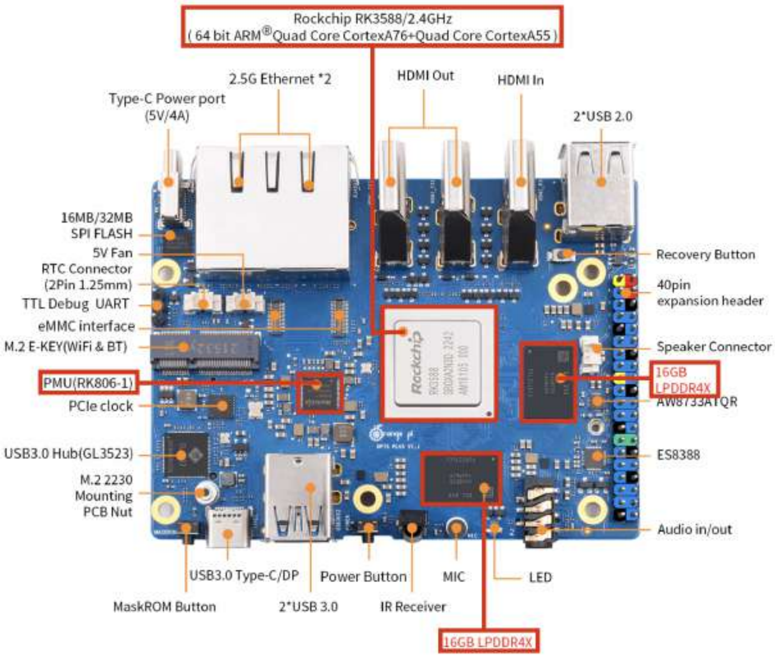
Heat sinks are devices used to dissipate heat generated by electronic components, ensuring they operate within safe temperature limits.

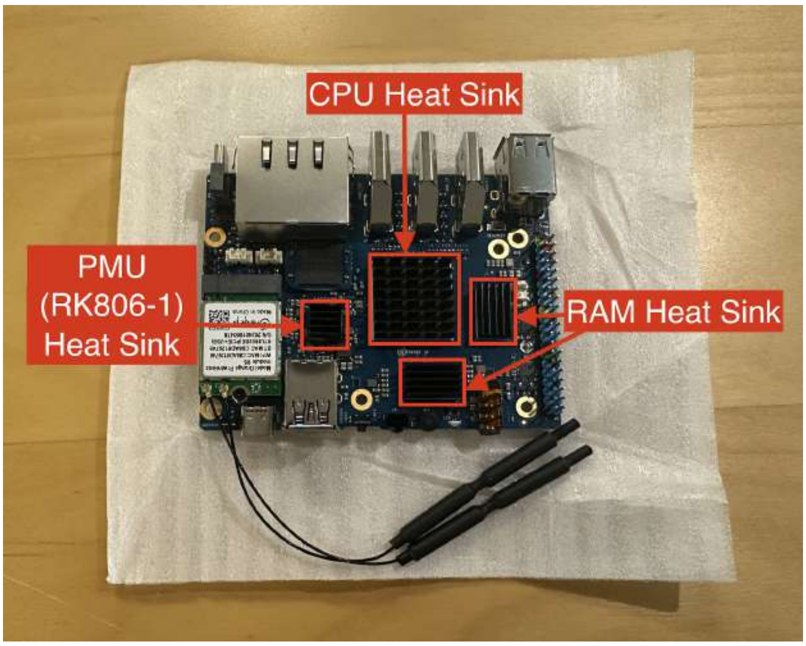
Why are Heat Sinks important?

We are utilizing these to ensure the CPU and memory components remain safe under high temperatures. This will aid in the device running smoothly.

Install Heat Sinks to required locations:

- The CPU requires the largest Heat Sink.
- The next two largest Heat Sinks are to be installed on the RAM.
- The last Heat Sink is to be installed on the PMU.





Step 3 - Connect WiFi Card

What is a WiFi Card?

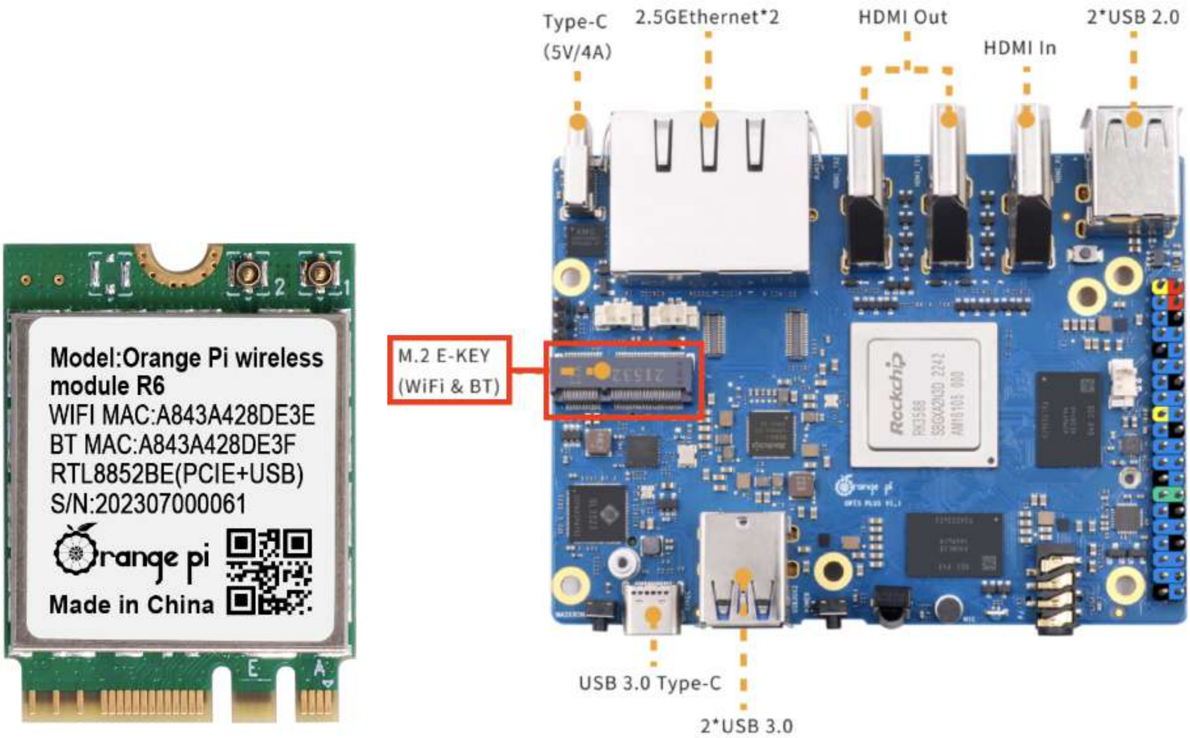
A WiFi card is a hardware component that enables a device to connect to a wireless network. It can come as an integrated component or as an add-on module that can be installed in a device.

Why is a WiFi Card important?

The WiFi card does not work currently; however, we are preinstalling this component with the anticipation that drivers will be released in the near future.

Connect the WiFi Chip to the Wi-Fi6/BT Module:

- Refer to the figure below for the install location of the Wi-Fi6/BT Module.



Step 4 - Install NVME

What is an NVME SSD?

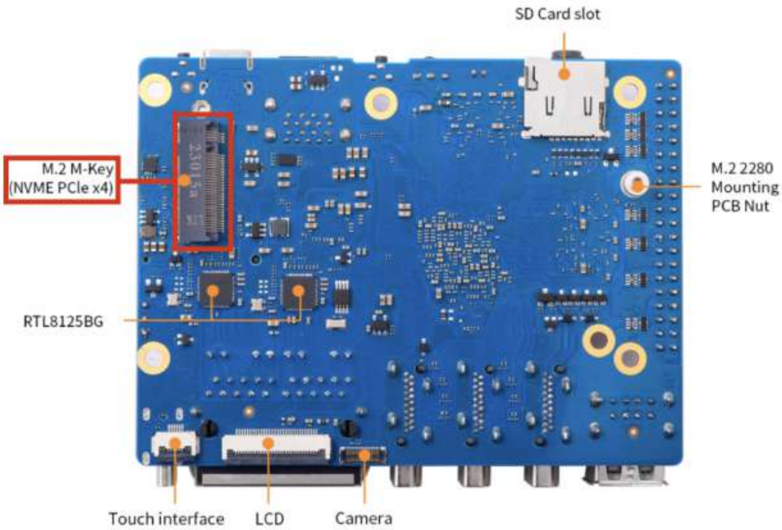
An NVME (Non-Volatile Memory Express) SSD is a high-speed storage device designed to take advantage of the fast data transfer speeds of the PCIe bus.

Why is an NVME SSD important?

Using an NVME SSD can significantly speed up the boot time and overall performance of the Orange Pi 5 Plus.

Install the NVME M.2 2TB SSD:

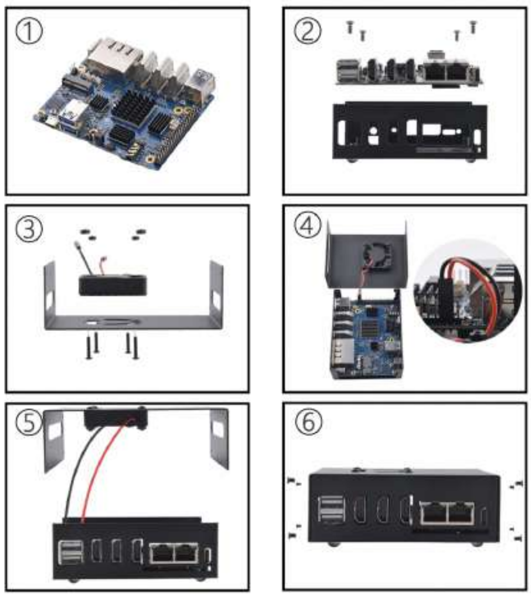
- Note: Additional M.2 screws are required. The 990 NVME M.2 2TB SSD does not include the screws.
- This is located on the back of the motherboard.



Step 5 - Mount Pi to Case

Mount Orange Pi 5 Plus to the bottom of the Metal Case:

- Orient the motherboard so that the ports on the motherboard line up with the slots on the case.
- Use the provided silver screws to mount the board into the case.
- Refer to this website for better visuals.
- Note: The fan should already be mounted on the top cover, if it's not, then follow the steps depicted in the figure below.



Step 6 - Install Fan

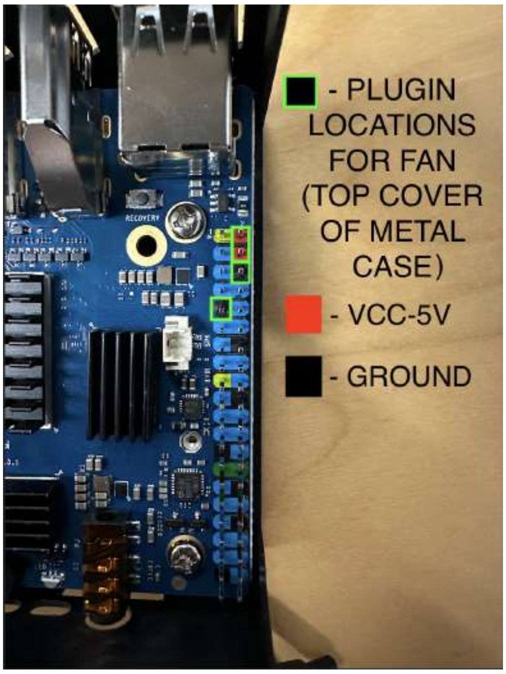
Install and plug in the fan from the top cover of the metal case:

- There are two cables: RED and BLACK.
- The BLACK cable gets plugged into the GROUND.
- The RED cables can be plugged into either of the VCC-5V plugs.
- Refer to this website for better visuals.


Pin definition

Orange Pi PC Plus GPIO Header

Pin#	NAME	NAME	Pin#
01	VCC-3V3	VCC-5V	02
03	TWI0-SDA (PA12)	VCC-5V	04
05	TWI0-SCK (PA11)	GND	06
07	PWM1 (PA6)	(PA13) UART3_TX	08
09	GND	(PA14) UART3_RX	10
11	UART2_RX (PA1)	(PD14) PD14	12
13	UART2_TX (PA0)	GND	14
15	UART2_CTS (PA3)	(PC4) PC4	16
17	VCC-3V3	(PC7) CAN_RX	18
19	SPI0_MOSI (PC0)	GND	20
21	SPI0_MISO (PC1)	(PA2) UART2_RTS	22
23	SPI0_CLK (PC2)	(PC3) SPI0_CS0	24
25	GND	(PA21) PA21	26
27	TWI1-SDA (PA19)	(PA18) TWI1-SCK	28
29	PA7 (PA7)	GND	30
31	PA8 (PA8)	(PG8) UART1_RTS	32
33	PA9 (PA9)	GND	34
35	PA10 (PA10)	(PG9) UART1_CTS	36
37	PA20 (PA20)	(PG6) UART1_TX	38
39	GND	(PG7) UART1_RX	40



- Below is a completed fan plugin orientation:

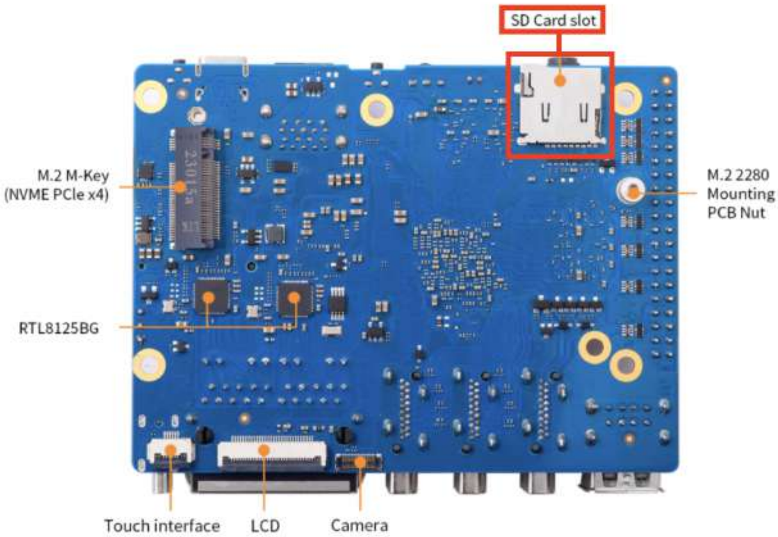
Step 7 - Ensuring Continuity

- If you plan to use more than one Orange Pi drone, one thing that might be useful is to label each of the drones with the hostname. This will allow you to figure out which drone is located where if issues arise.

Startup

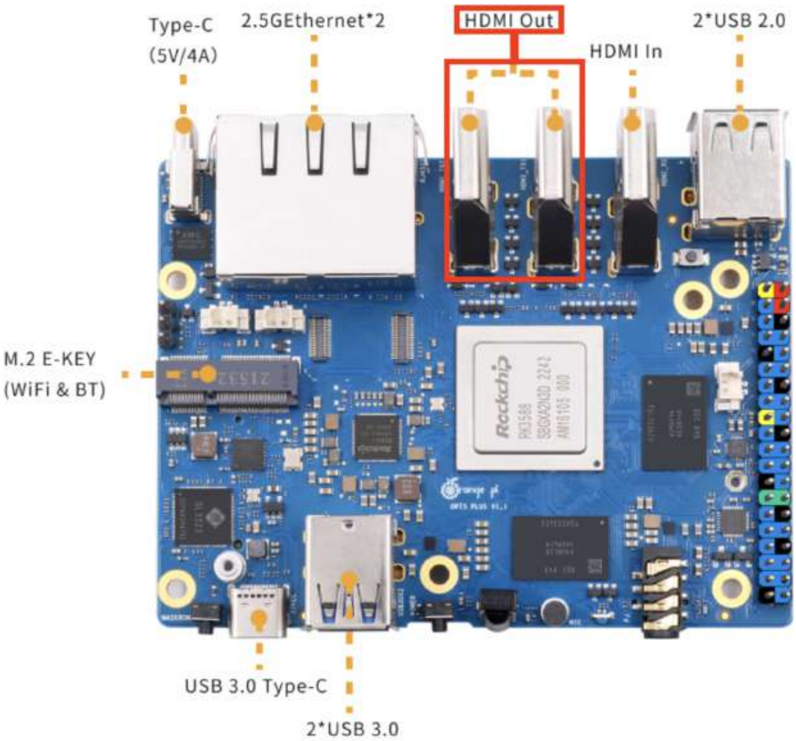
Step 1 - Load SD Card

- Plug the SD Card in the front SD Card slot reader. This is located on the side with the lights and blue USB ports.



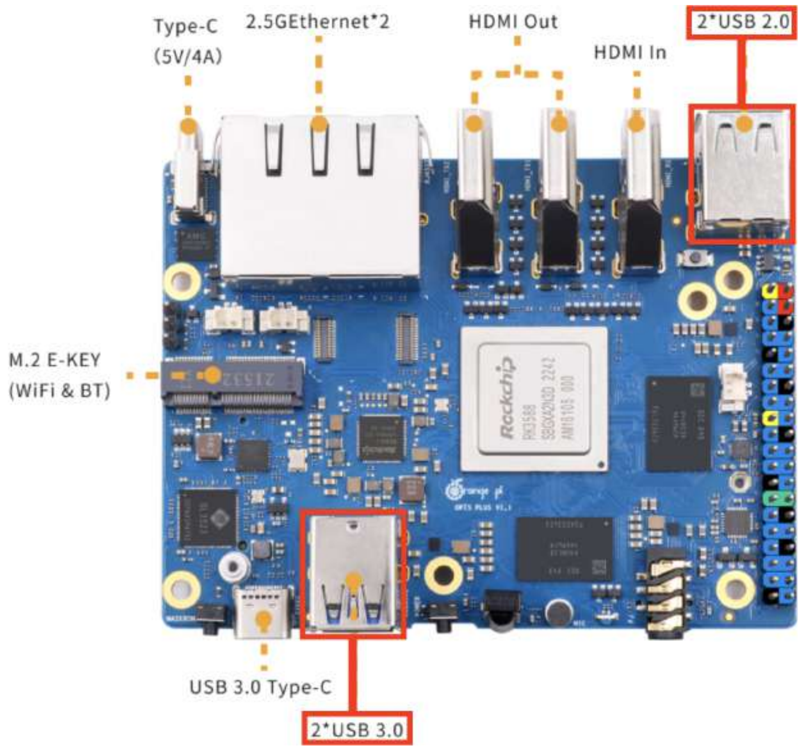
Step 2 - Connect Visuals

- Plug the HDMI cable from a monitor in either of the HDMI Out slots closest to the ethernet ports.



Step 3 - Connect Peripherals

- Plug the USB hub into a USB port on either side of the device.



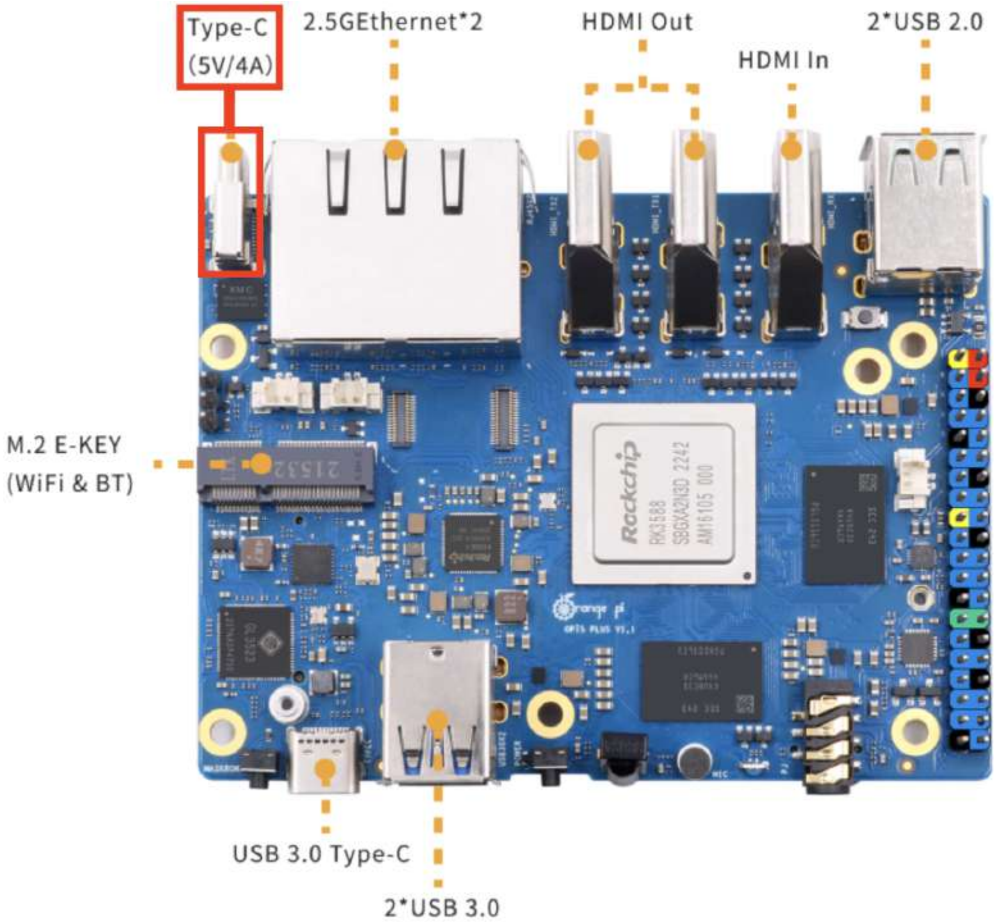
Step 4 - Connect Peripherals Part 2

- Connect the USB keyboard and mouse to the USB hub.

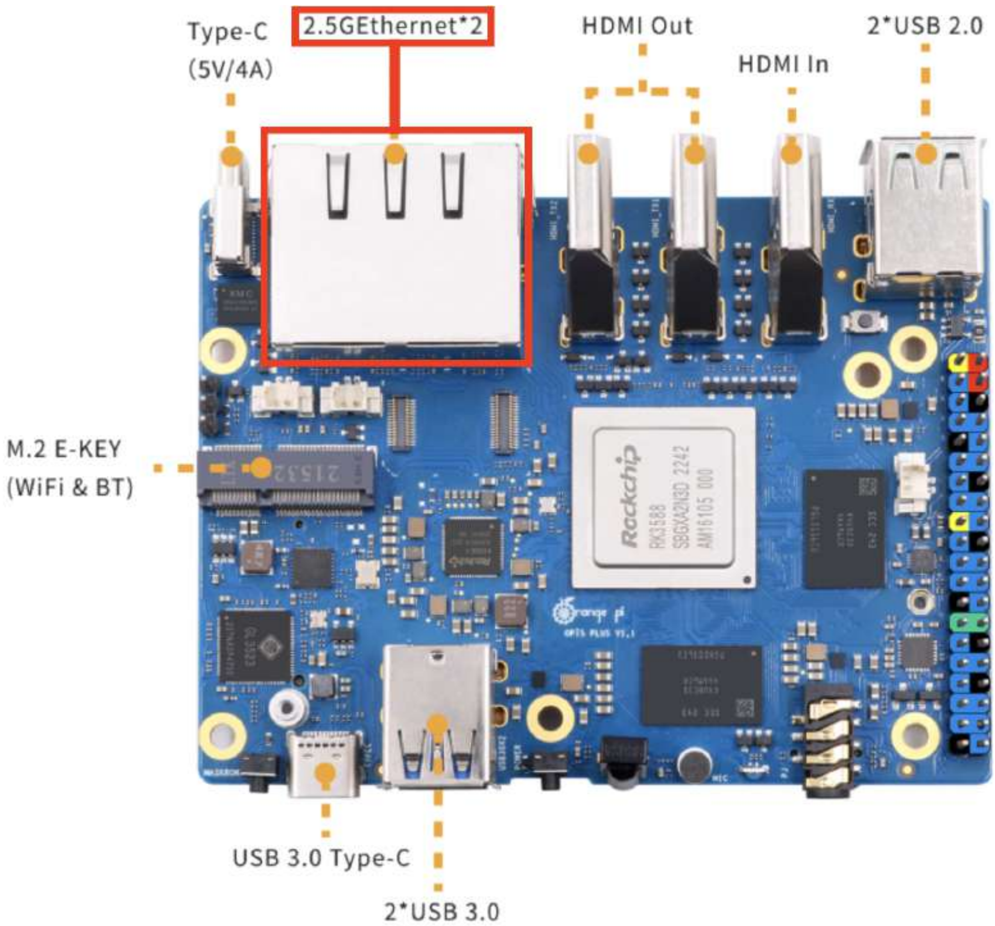
Step 5 - Add Power and Data Linkage

- Plug the ethernet cable into the PoE (Power Over Ethernet) splitter.

- Plug the Type-C cable from the PoE into the Type-C 5V/4A slot.



- Then plug the ethernet cable from the PoE into one of the ethernet ports next to the Type-C slot.



- The Orange Pi should boot and display on the monitor after plugging in the Type-C cable from the PoE.
- If the Orange Pi does not boot, then power-cycle the device (If the LED is purple/pink).
    - To power cycle:
        - Unplug the PoE Type-C cable.
        - Wait about 15-30 seconds.
        - Plug the PoE Type-C back in.
- If the LED is flashing then it is reading the disk and all is good.
- To turn off the device, unplug the Type-C cable.

Step 6 - Flash the eMMC

Open the terminal to install the Operating System onto the eMMC:

- Command:

```
sudo nand-sata-install
```

- Select option #2 – Boot from eMMC – system on eMMC.
- Click Yes

- Click option #1 – ext4
    - If it says “Partition Too Small” --> go to cfdisk /dev/mmcblk0 --> delete all partitions.
- Click “Power Off” then immediately remove the SD Card.

Step 7 - Power On Orange Pi 5 Plus

- To power the device back on click the button and you should see a flashing light.
- This is located on the side with the 2 (Blue) USB 3.0 ports.
- The OS should boot from the eMMC.
- Note: A user should already be created (user1).

Step 8 - Connect ALFA WiFi Card:

- Connect the ALFA WiFi Card via USB 3.0 to either of the front USB ports (Blue).
- Command to check if the interface is recognized:

```
ip a
```

- The interface should look something like this:

```
wlx00**********
```

Step 9 - Assign Host Names:

- Navigate to Settings --> System --> About.
- Change the “Device Name” to be whatever you want.
    - EX: drone1
- Next, navigate to the terminal and grab the MAC Address.
    - Command:

    ```
    ip a
    ```

- Locate the correct ethernet interface:

```
enP******
```

- Navigate to the /etc/hosts file:
    - Commands:

      ```
      cd /
      sudo nano /etc/hosts
      ```

    - Locate “127.0.1.1” and change “orangepi5-plus” to the new drone's name.
        - Before Change:
            - orangepi5-plus
        - After Change:
            - drone01
    - Locate “::1” and replace “orangepi5-plus” with the new drone's name.
        - Before Change:

          ```
          localhost orangepi5-plus ip6-localhost ip6-loopback
          ```

        - After Change:

          ```
          localhost drone01 ip6-localhost ip6-loopback
          ```

- Reboot to test the configuration has been set.

Kismet

Step 1 - Kismet Server Configuration Setup

Log Into The Central Kimset Server

- Commands:

```
ssh username@SERVERIPADDRESS
```

Navigate To The Kismet Configuration File

- Commands:

```
cd /usr/local/etc
sudo nano kismet.conf
```

- Navigate to lines #35 - #37 and edit the server_name, server_description, and server_location to your liking.

```
# Kismet can report basic server information in the status response, this
# can be used in some situations where you are running multiple Kismet
# instances.
#
server_name=Kismet
server_description=A Kismet server.
server_location=Your Favorite Location
```

- Next, navigate to line #91 and ensure that the remote_capture_enabled is set to true. Then ensure that remote_capture_listen is = to 0.0.0.0. Finally, leave the port as is.

```
# Kismet can accept connections from remote capture datasources; by default this
# is enabled on the loopback interface *only*.  It's recommended that the remote
# capture socket stay bound to the loopback local interface, and additional
# authentication - such as SSH tunnels - is used.  Check the Kismet README for
# more information about setting up remote capture securely!
#
# Remote capture can be completely disabled with remote_capture_enabled=false
remote_capture_enabled=true
remote_capture_listen=0.0.0.0
remote_capture_port=3501
```

- Lastly, navigate to line #133 and add the following command under the comment:
    - Ensure that you change the "IPAddress" to a static IP for easy tracking in the future.

```
# Kismet does not pre-define any sources, permanent sources can be added here
# or in kismet_site.conf
ncsource=drone:tcp:IPADDRESS:3501
```

- Note: Alec assigned static IP Addresses for each of the drones. This will be helpful for easy tracking in the future.

Step 2 - Locate Interface

Determine The Correct Interface For WiFi Capture & RTL-SDR Capture

- On the Orange Pi 5 Plus drone navigate to the terminal and type the following command:
    - Command:

    ```
    ip a
    ```

- Locate either the WiFi capture interface or the RTL-SDR interface.

Step 3 - Set Aliases

Set Aliases For Kismet Capture Types On The Orange Pi 5 Plus Drone

- Alias the commands for WiFi or RTL-SDR as “kismetwificap” and “kismetdrcap”:
    - Commands:
        - Setup command for WiFi capture using an ALFA WiFi Card:

        ```
        alias kismetwificap='kismet_cap_linux_wifi --connect IP ADDRESS OF KISMET SERVER:PORT --tcp --source wlx00********** &'
        ```

        - Setup command for RTL-SDR capture using a Nooelec RTL-SDR:

        ```
        alias kismetdrcap='kismet_cap_sdr_rtl433 --tcp --connect IP ADDRESS OF KISMET SERVER:PORT --source rtl***-* &'
        ```

        - If you want to start Kismet and keep it running even if the shell is closed, add "nohup" to the beginning of the alias's:
            - WiFi Capture:

```
alias kismetwificap='nohup kismet_cap_linux_wifi --
connect IP ADDRESS OF KISMET SERVER:PORT --tcp --source
wlx00********** &'
```

- RTL-SDR:

```
alias kismetdrcap='nohup kismet_cap_sdr_rtl433 --tcp --
connect IP ADDRESS OF KISMET SERVER:PORT --source
rtl***-* &'
```

- If ever you need to change the alias, this is located in the .bashrc file under the home directory:

```
cd ~
sudo nano .bashrc
```

- Note: The source for the wificap changes based on the MAC Address of the wireless card.

- Next, type the following command to create a new shell instance with the updated .bashrc file:

```
bash
```

Step 4 - Start Kismet

- Command to start WiFi capture:

```
kismetwificap
```

- Command to start RTL-SDR capture:

```
kismetdrcap
```

Step 5 - System Test

- Navigate to your Kismet web interface.

- Select hamburger menu on the left hand side.
- Navigate to Data Sources.
- You should see the interface for the service you are running.
    - EX: WiFi capture interface (wlx00**********)
    - EX: RTL-SDR capture interface (rtl***-*)

Metricbeat

Step 1 - Install Metricbeat

- Command to install Metricbeat on the Orange Pi:

```
curl -L -O
https://artifacts.elastic.co/downloads/beats/metricbeat/metricbeat-
8.14.1-arm64.deb
sudo dpkg -i metricbeat-8.14.1-arm64.deb
```

- Note: The Metricbeat download version (8.14.1) should be the same as your ELK Stack version.

Step 2 - Edit The Metricbeat Configuration File

- Navigate to the Metricbeat configuration file:

```
sudo nano /etc/metricbeat/metricbeat.yml
```

- Under setup.kibana in the Kibana section, uncomment and change the command below:
    - Before Change:

    ```
    host: "localhost:5601"
    ```

    - After Change:

    ```
    host: "ELKSERVERIPADDRESS:5601"
    ```

- Under output.elasticsearch in the Outputs section, change hosts:
    - Before Change:

    ```
    host: ["localhost:9200"]
    ```

- After Change:

  ```
  host: ["ELKSERVERIPADDRESS:9200"]
  ```

- Next, uncomment the following line:

```
protol: "https"
```

- Uncomment the following lines after the "# Authentication credentials":

```
username: "ELKWEBINTERFACEUSERNAME"
password: "ELKWEBINTERFACEPASSWORD"
```

- Add the following lines after the the previously uncommented lines:

```
ssl.verification_mode: "none"
```

Step 3 - Finalize Setup

- To finish the setup:

```
sudo metricbeat setup -e
```

Step 4 - Start Metricbeat

- To start metricbeat:

```
sudo service metricbeat start
```

Step 5 - Enable Service

- To enable the Metricbeat service, run the following command:

```
sudo systemctl enable metricbeat
```

Step 6 - System Test

- Navigate to your Kibana web interface.
- Select observability.
- The drone host-name should appear in the "Hosts" dropdown menu if everything was done correctly.